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SECJ2013 - DSA

2024/2025-1

QUIZ 4

Queue

Given a structure of basic queue (`myQueue`) implementation by using linear array as follows:

	front		*back*	
0	**1**	**2**	**3**	**4**
50	20	30	24	

1. How many items currently exist inside the queue?___3____

2. Draw the queue structure after the following instructions were executed:

```
myQueue.dequeue();
myQueue.dequeue();
myQueue.enqueue(60);
myQueue.enqueue(47);
```

	front		*back*	
0	**1**	**2**	**3**	**4**
50	20	30	24	

3. Based on the same instructions given in question 2 above, draw the queue structure if the circular-queue technique is implemented:

	front		*back*	
0	**1**	**2**	**3**	**4**
50	20	30	24	

2. Draw the queue structure after the following instructions were executed:

```
myQueue.dequeue();
myQueue.dequeue();
myQueue.enqueue(60);
myQueue.enqueue(47);
```

```
Initial:
```

	front		*back*	
0	**1**	**2**	**3**	**4**
50	20	30	24	

```
myQueue.dequeue();
```

		front	*back*	
0	**1**	**2**	**3**	**4**
50		30	24	

```
myQueue.dequeue();
```

			front back	
0	**1**	**2**	**3**	**4**
50			24	

```
myQueue.enqueue(60);
```

			front	*back*
0	**1**	**2**	**3**	**4**
50			24	60

```
myQueue.enqueue(47);
```

			front	*back*
0	**1**	**2**	**3**	**4**
50			24	60

3. Based on the same instructions given in question 2 above, draw the queue structure if the circular-queue technique is implemented:

```
myQueue.dequeue();
myQueue.dequeue();
myQueue.enqueue(60);
myQueue.enqueue(47);
```

```
Initial:
```

front *back*

0	1	2	3	4
50	20	30	24	

```
myQueue.dequeue();
```

front *back*

0	1	2	3	4
50		30	24	

```
myQueue.dequeue();
```

front back

0	1	2	3	4
50			24	

```
myQueue.enqueue(60);
```

front *back*

0	1	2	3	4
50			24	60

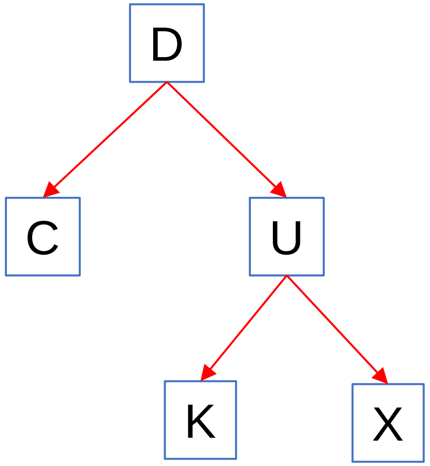
```
myQueue.enqueue(47);
```

back *front*

0	1	2	3	4
47			24	60

Tree

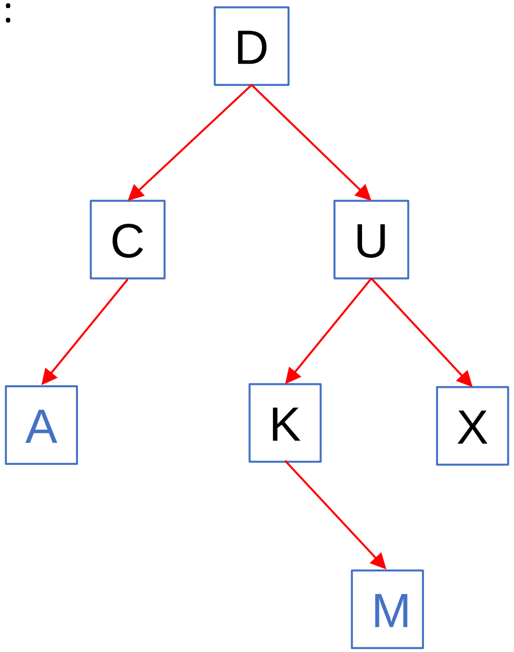
Given a binary tree structure as follows:



1. Trace the output for pre-order, in-order, and post-order traversals

Pre-order: D C U K X (top-left-right)
In-order: C D K U X (left-top-right)
Post-order: C K X U D (left-right-top)

2. Draw the new structure of the tree after new nodes/items (‘A’ and ‘M’) are added into the tree:

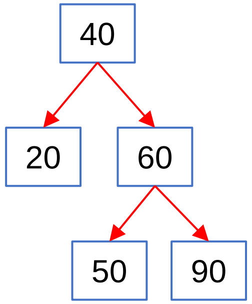


A<D → left child (C)
A<C → left child

M>D → right child (U)
M<U → left child (U)
M>K → right child

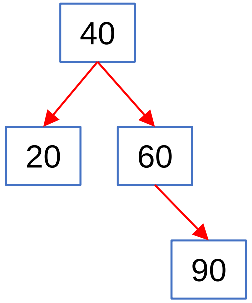
Tree

Given a binary tree structure as follows:

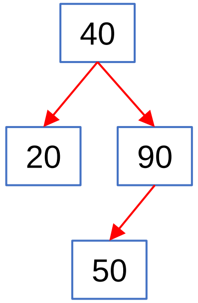


For each question below, assume that the tree structure is always in its original form, as shown on the left side of this slide page.

1. Tree structure after node 50 is deleted:



2. Tree structure after node 60 is deleted:



3. Tree structure after node 40 is deleted: